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The effects of micro-aeration on semi-continued anaerobic digestion of corn straw with increasing organic loading rates

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ABSTRACT

The benefits of micro-aeration on batch anaerobic digestion (AD) of organic wastes have been intensively studied. However, the effects of micro-aeration on the semi-continued AD under different organic loading rates (OLRs) were still lacking. This study investigated the effects of micro-aeration on semi-continued AD of corn straw with increasing OLRs from 1.1 to 2.1 volatile solid (VS)/(L·day). Results showed micro-aeration could benefit the AD process by increasing the median daily methane yields for 6%, 10% and 8% at OLRs of 1.1, 1.5 and 2.1 VS/(L·day), respectively. Moreover, micro-aeration stimulated the growth of facultative bacteria (*Clostridia*), which secreted more critical enzymes (e.g., cellulase and xylanase) and led faster degradation of corn straw as well as higher soluble humic acids (SHAs) concentration. The SHAs might act as electron shuttle to facilitate the direct interspecies electron transfer (DIET) among syntrophic bacteria (*Cloacimonadia*) and hydrogenotrophic methanogens (*Methanobacterium*), which led to a faster volatile fatty acids (VFAs) conversion into methane. This study confirmed the beneficial effects of micro-aeration on the semi-continued AD at different OLRs and unraveled the mechanisms behind.

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1. Introduction

Anaerobic digestion (AD) is a vivid technology in renewable energy production, waste streams stabilization, and greenhouse emissions reduction [1]. However, the applications of AD technology often meet many barriers such as long fermentation time, low substrate conversion efficiencies, and vulnerability to the impacts of toxic or high organic loading rate (OLR) [2–4]. The reason for the barriers mentioned above could be explained by the inherent difficulty in maintaining a metabolism balance among series of microorganisms (i.e., hydrolytic bacteria, acidogens, acetogens, and methanogens) [5]. Moreover, the interactions between these microbial communities are still poorly unraveled [6]. Managing or altering the microbial community compositions to achieve specific goals that is the so-called microbial resource management has been considered as an efficient way to influence the outcome of the AD process [7]. Microbial resource management has been also proved to be efficient in increasing the tolerance of AD system to ammonia [8], lipid-rich substrate [9], as well as in enhancement of the lignocellulosic materials biodegradation [10,11], etc.

Oxygen is traditionally considered as an inhibiting factor for AD process [5]. However, recent studies have proved a little breath of oxygen or air introduced into the AD process (that is so called micro-aeration) could benefit the AD performance by regulation the microbial community composition [5,12,13]. For example, Fu et al. [12], found that micro-aeration at the intensity of 12.5 ml air/$L_{reactor}$ per day led 16.5% higher methane yield from AD of corn straw. The methanogenesis and hydrolysis efficiency of blackwater

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under micro-aeration condition (5 mg $O_2/L_{reactor}$ per cycle) increased from 39.6% to 50.7% and from 34.7% to 48.7%, respectively [14]. The total VFAs concentration rapidly decreased and its inhibition effect on methanogens was alleviated when ORP-controlled micro-aeration was adopted during AD of Napier grass at high organic loading rate [5]. As reported in previous studies, micro-aeration could stimulate growth of facultative bacteria, which would produce more extracellular enzyme in turn, and promote hydrolysis rate as well as VFAs production [12,15,16]. Nevertheless, these studies were mainly conducted in batch experiment and low organic loading rates. The effects of micro-aeration on semi-continued AD of cellulosic substrate are rarely reported, the mechanisms behind are also not fully revealed though it is vital to AD process.

The aim of this study was to develop and validate the beneficial effects of micro-aeration on the semi-continued AD of corn straw under increasing OLRs from 1.1 to 2.1 g VS/(L day). The mechanisms behind were comprehensively analyzed according the reactor performance, bacterial and archaeal community, extracellular enzyme, and VFAs consumption etc. The implement of this study could further deepen the understanding of micro-aeration and provide references for its industrial application.

2. Material and methods

2.1. Substrate and inoculum

Corn straw, a typical biomass waste in agriculture, was selected as a representative lignocellulosic substrate in this paper. The biomass was harvested from a corn farm (Xuzhou, Jiangsu Province, China). The freshly corn straw was air-dried to the moisture content about 6.6%, and then milled by a cutting mill (BJ-150, Baijjie, Shanghai, China) to approximate particle size of 0.1 cm, which was then stored in plastic vacuum bags for further use. The total solids (TS), volatile solids (VS), content of hemicellulose, cellulose and lignin of the corn straw were 93.4 ± 0.2%, 90.7 ± 0.2% (% of TS), 19.5 ± 4.6% (% of TS), 34.0 ± 2.0% (% of TS) and 22.7% ± 0.7% (% of TS), respectively.

The inoculum used was granular anaerobic sludge that collected from a mesophilic up-flow anaerobic sludge blanket (UASB) reactor treating citric acid wastewater. Before usage, the inoculum was homogenize using a grinder (MJ-BL25B3, Midea, Guangdong, China) for 3 min to obtain homogeneous liquid. The TS and VS of the inoculum were 36.4 ± 3.7% and 34.6 ± 1.3% (% of TS).

2.2. Anaerobic reactor set-up and operation

The design of anaerobic reactor was shown in Fig. S1. Duplicate semi-continued AD experiments were carried out in double-layer glass reactors with working volume of 2 L. The hydraulic retention time (HRT) of reactors was set as 33 days, and the reactors were running at 37 ± 1 °C with water bath (HH–S24s, Dadi, Jintan, China) heat preservation. The biogas produced was collected using 4 L gas sampling bags (MBT32-4, Hedetech, Jiangsu, China). The organic loading rates (OLRs) were increased gradually from 1.1 g VS/(L·day) at the 1–118 days to 1.5 g VS/(L·day) at the 119–178 days, and finally reached 2.1 g VS/(L·day) at the 179–238 days. The micro-aeration intensity was set as 0.2 mL/(g·VS) according to our previous study [13], which was realized by pure oxygen (>99.98%) injection from the bottom tube using a micro-peristaltic pump (BT100-1L, LongerPump, Heibei, China) at the speed of 3.75 mL/h. Based on the time of micro-aeration and OLRs, the two groups were divided into 4 stages: Stage1, 1–50 days with the OLRs of 1.1 g VS/(L·day) named C0 (without micro-aeration) and M0 (without micro-aeration); Stage 2, 51–118 days with the OLRs of 1.1 g VS/(L·day) named C1.1 (without micro-aeration) and M1.1 (micro-aeration); Stage3, 119–178 days with the OLRs of 1.5 g VS/(L·day) named C1.5 (without micro-aeration) and M1.5 (micro-aeration); Stage 4, 179–238 days with the OLRs of 2.1 g VS/(L·day) named C2.1 (without micro-aeration) and M2.1 (micro-aeration). For the start-up of AD, nutrient solution [17] was used to adjust the working volume to 2 L in each reactor, and the water discharged with digestate was refilled with tap water.

2.3. Effects of different concentrations of extracted SHAs on methanogenesis

Effects of different concentrations of extracted SHAs on methanogensis were explored in batch model using 4 g/L sodium acetate as the sole substrate. Concentrations of SHAs were set as 0 (control), 0.5(SHAs0.5), 1 (SHAs1), and 2 (SHAs2) g/L, respectively. 60 mL inoculum from the long-time running reactors was collected at the 220th day, which was then mixed with nutrient solution [17] to the total liquid volume of 200 mL. Syringe bottles were purged with nitrogen for 5 min to create anaerobic condition, sealed with rubber stoppers and then incubated in a 37 °C air bath at 200 rpm. The biogas was measured in volume and composition every day. The modified *Gompertz* model [18] was also used to fit the cumulative methane yield to analyze the effect of HAs concentration on the methane productions (Supplementary Information–Supporting Methods).

2.4. Microbial community structure analysis

At the 118th, 178th and 238th day of AD, 10 mL of biogas slurry were collected from control and micro-aeration reactors for microbial community structure analysis as the method described in our previous study [19]. Briefly, DNA was firstly extracted from collected samples, and then the high throughput sequencing method was used to analyze the microbial community structure. The primer set for the V3–V4 region amplification of the bacterial 16S rRNA gene was 338F (ACTCCTACGGGAGGCAGCAG) and 806R (GGACTACHVGGGTWTCTAAT), while the primer set for the V4–V5 region amplification archaeal 16S rRNA gene was 524F (TGYCAGCCGCCGCGGTAA) and 958R (YCCGGCGTTGAVTCCAATT). The gene data was processed on Illumina Miseq sequencing platform (Majorbio, Shanghai, China) for the microbial diversity analysis. In order to investigate the effects of micro-aeration on the extracellular enzymes during AD process, PICRUSt, a software for the prediction of genome functions, was used to predict the abundances of hydrolytic enzymes by the online platform (Majorbio, Shanghai, China) as described in a previous study [20].

2.5. Analytical methods

2.5.1. Analysis of soluble humic acids in digestate

Digestate for SHAs analysis were collected at 118th (C1.1 and M1.1), 178th (C1.5 and M1.5) and 238th (C2.1 and M2.1) days. EEM fluorescence spectra was conducted to investigate the effects of micro-aeration on SHAs in the collected digestate during the semi-continued AD (Supplementary Information–Supporting Methods). In order to study the effects of micro-aeration on the concentration of SHAs, SHAs were extracted according to the procedure developed by the International Humic Substances Society (IHSS), and the gravimetric quantification of the SHAs in g/L were conducted according to the details in the Supplementary Information-Supporting Methods. The surface functional groups of SHAs precipitates were also analyzed through Fourier transform infrared (FTIR) spectroscopy (IRTracer-100, Shimadzu, Japan).

2.5.2. Cyclic voltammetry (CV)

CV is a useful electrochemical method to analyze the redox properties of materials and bio-system [21]. In order to study the effects of micro-aeration on the redox properties of AD process, digestate collected from control and micro-aeration reactors at the end of experiment (238th day) were tested by a three electrode electrochemical cell in a single chamber (Supplementary Information -Supporting Methods).

The effects of extracted HAs on CV of digestate were also conducted as follows: four solutions were prepared: 200 mL nutrient solution [17], 200 mL nutrient solution +0.2 g extracted SHAs, 100 mL nutrient solution + 100 mL digestate, and 100 mL nutrient solution + 100 mL digestate + 0.2 g extracted SHAs, namely, Nutrient, Nutrient + SHAs, Nutrient + Digestate, and Nutrient + Digestate + SHAs, respectively.

2.5.3. Other analysis

The volume and composition of biogas, pH and VFAs concentration were determined as the procedures described in our previous study [13].The TS and VS were determined according to the gravimetric method [22]. Lignocellulosic compositions of corn straw were analyzed according to the procedure developed by National Renewable Energy Laboratory (NREL) [23].

3. Results and discussion

3.1. Methane production from semi-continued anaerobic digestion corn straw

Fig. 1a shows the daily methane yields during the AD of corn straw under different OLRs in control and micro-aeration reactors. The control and micro-aeration group possessed the same trend. The daily methane yields in control and micro-aeration groups were unstable until 50 days after start-up, then the reactors reached a relatively stable status. When oxygen was beginning to be injected into the micro-aeration reactors at 50th of AD, the daily methane yield of the micro-aeration group was firstly negatively affected by oxygen injection, and showed lower methane yield than control from 51st to 77th d. Afterwards, the daily methane yield in micro-aeration group was higher than control until the end of experiment. The results indicated that the suddenly addition of oxygen might interrupt the methane fermentation process and the microorganisms in reactors could accommodate the micro-aeration circumstances rapidly, which reflected as the increased methane production.

In order to further analyzing the effect of micro-aeration on the AD of corn straw under different OLRs, the daily methane yields during the final 30 days (when the reactors were running stably) of each period were selected to represent the daily methane yield variations and analyzed by violin with box plots, as shown in Fig. 1b. The black box and line whisker in both control and micro-aeration groups became shorter with the increase of operation time, but the median methane yields in two group were different. The median daily methane yields in C0 and M0 were 231.7 ± 15.1 and 232.4 ± 10.2 NmL/(g·VS·d), indicating the successful start-up of the control and micro-aeration groups. The median daily methane yields in M1.1, M1.5 and M2.1 were 213.5 ± 9.4, 164.8 ± 5.5 and 152.9 ± 2.8 NmL/(g·VS·d), which were 6%, 10% and 8% higher than those in C1.1, C1.5 and C2.1, respectively. Due to the complex components and recalcitrant structure in lignocellulosic biomass, the hydrolysis was considered as the rate-limiting process for AD [5,13]. Micro-aeration has been reported to promote methane yield by accelerating hydrolysis of AD in several studies [12,15,16]. The results demonstrated that micro-aeration could lead higher methane yields at semi-continued AD reactors with different organic loads.

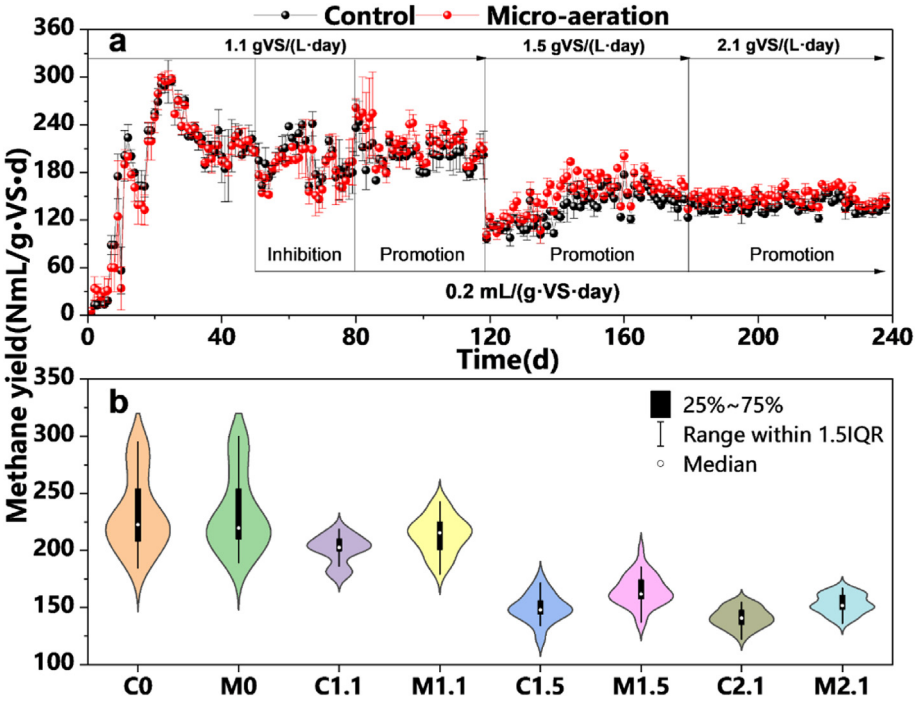


Fig. 1. Daily methane yield (a) and violin with box plot for daily methane yield from the final 30 days of anaerobic digestion of corn straw under different OLRs in control and micro-aerobic reactors (b). For violin with box plot, the width of violin with box plot indicates the frequency of the daily methane yield, the top and bottom box are the 75th and 25th percentile of the group, the higher and lower whisker represents the range within 1.5 IQR of the group, while white small circle shows the median daily methane yield of the group.

3.2. VFAs concentration variations under different OLRs in control and micro-aeration reactors

In order to investigate the effects of micro-aeration on VFAs productions during AD of corn straw under different OLRs, VFAs concentrations in digestate were tested when the reactors were running stably. The VFAs concentration variations under different OLRs in control and micro-aerobic reactors are shown in Fig. 2. During the experiment, acetate and propionate were the main compositions of VFAs, while the concentrations of other volatile fatty acids (e.g., butyrate) below the detectable limit. Similar results were also reported that VFAs were only composited by acetate (0.15 ± 0.22 g/L) and propionate (0.15 ± 0.25 g/L) during the semi-continued AD of Napier grass at the OLR of 1.5 g VS/(L·day) [5]. VFAs concentrations increased along with the increasing of OLRs. For instance, at the final day of each OLRs, total VFAs concentrations in C1.1(118th d), C1.5(177th d) and C2.1 (238th d) were 0.41 ± 0.02, 0.53 ± 0.15 and 0.67 ± 0.18 g/L, respectively. Relatively, the total VFAs concentrations in M1.1 (118th d), M1.5 (178th d) and M2.1 (238th d) were 0.49 ± 0.02, 0.65 ± 0.31 and 0.91 ± 0.30 g/L, which were 20.0%, 45.7% and 35.2% higher than those in C1.1, C1.5 and C2.1, respectively. The results reflected micro-aeration could enhance the hydrolysis process of corn straw, which led increased VFAs concentration.

3.3. Soluble humic acids released under different organic load rates

Lignocellulosic biomass like corn straw is composed by cellulose, hemicellulose and lignin according non-covalent forces and covalent cross-linkage [24]. Limited by the dense structure, hydrolysis step is considered as the rate-limiting step during AD of corn straw [12]. To examine whether micro-aeration treatment caused faster degradation of corn straw, EEM technique was applied to analyze the biogas slurry obtained from different OLRs in control and micro-aeration groups, which are shown in Fig. 3. Fluorescence responses were weak in Regions I, II, III, and IV, while the fluorescence responses in Region V (represented humic acid-like organics) were strong. Furthermore, fluorescence responses in Region V became stronger with micro-aeration and the increase of OLRs. Consistent with the EEM results, the SHAs concentrations in M1.1, M1.5 and M2.1 were 1.6 ± 0.1, 1.8 ± 0.1 and 2.1 ± 0.1 g/L, which were 60.0%, 38.5% and 50.0% higher than those in C1.1, C1.5 and C2.1, respectively (shown in Fig. 4). As one of the by-products during AD of cellulosic substrate, HAs are originated from the degradation of lignin, cellulose and other non-lignin substances through the metabolism of microorganisms [25]. Thus, higher degradation of corn straw might lead increased concentration of SHAs, which led to stronger fluorescence response in Region V in turn. The high degradation of corn straw could be further proved by the methane yield as shown in Table 1. The average daily methane yields from the final 30 days in C1.1, M1.1, C1.5, M1.5, C2.1 and M2.1 were 441.5 ± 26.7, 468.0 ± 20.7, 470.9 ± 20.0, 517.8 ± 17.6, 629.6 ± 22.4 and 674.5 ± 22.7 NmL/d, which showed increasing tendency with the micro-aeration and increased OLRs. The Max and Min daily methane yield from the final 30 days in C1.1, M1.1, C1.5, M1.5, C2.1 and M2.1 shared the same trends with average methane yields. Higher methane yield in micro-aeration and higher OLRs groups reflected that more substances (corn straw) were converted into methane, SHAs and other end-products by microorganisms in anaerobic digesters.

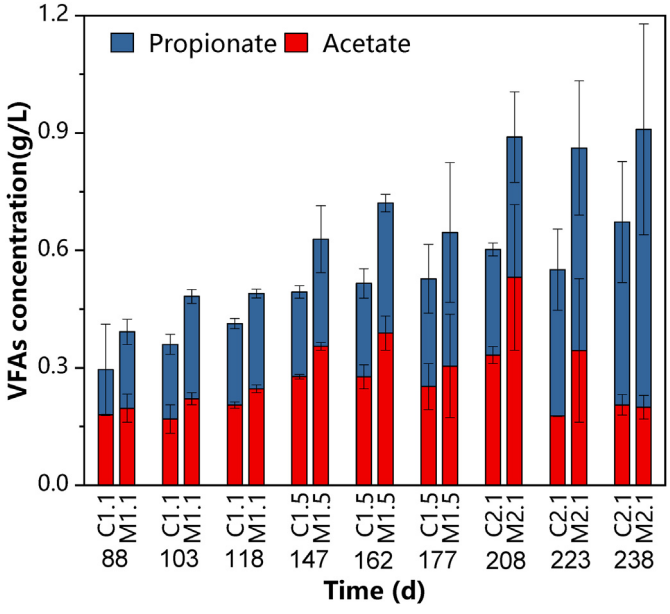


Fig. 2. VFAs concentration Variations under different OLRs in control and micro-aerobic reactors.

3.4. Cyclic voltammetry analysis of digestats and the digestate with HAs addition

AD is a process including a series of interdependent biochemical steps carried out by diverse microorganisms (i.e., hydrolytic bacteria, acidogens, acetogens, and methanogens), in which metabolic products like organics, hydrogen, carbon dioxide, electrons that excreted by one group microbes are essential substrates for next group of microbes [15]. Electrons flows among microorganisms are vital for the improvement of methane production and stabilization of AD process [21], and the electron transfer capacity can be analyzed by cyclic voltammetry method [26]. Fig. 5a shows the CV at the scan rate of 50 mV/s of digestate from C2.1 and M2.1. It could be observed that three distinct redox peaks with potentials of 0.68 (peak1), 0.28 (peak2), and −0.16 V (peak3) (versus saturated calomel electrode) presented in all two groups. The peak1 currents of M2.1 was 6.8 mA, which was 28.3% higher than that of C2.1, while peak2 and peak3 currents showed little differences between C2.1 and M2.1. The higher peak currents occurred in M2.1 demonstrated the higher capacity of electron transfer, which might attribute to the increased SHAs under micro-aeration condition. And the result was further confirmed by Fig. 5b, which showed that the addition of HAs promoted the peak currents in peak1 and peak2. As reported, the HAs could act as electron transport medium [27–29] and the higher SHAs concentration might lead higher electron transfer capacity to AD system, which reflected by the CV analysis. In addition, FTIR analysis in Fig. S2 showed that the SHAs extracted from digestate of AD of corn straw contained abundance of surface oxygen-containing functional groups like C—O, C═O, and C—O, which could serve as electron transport medium and led higher electron transfer capacity to AD system [21]. The results indicated that micro-aeration might create a favorable electron transfer environment for the metabolisms of syntrophic microorganisms.

3.5. Analysis of the microbial community structure

The bacterial community structures at class level are shown in Fig. 6a. The relative abundances of *Clostridia* in M1.1, M1.5 and M2.1 were 13.0%, 17.8% and 23.3%, which were 16%, 9.4% and 13.9% higher than those of C1.1, C1.5, and C2.1, respectively. *Clostridia* (phylum *Firmucutes*) are well known to play a vital role in the AD of organic waste for their ability to excrete extracellular enzyme to accelerate

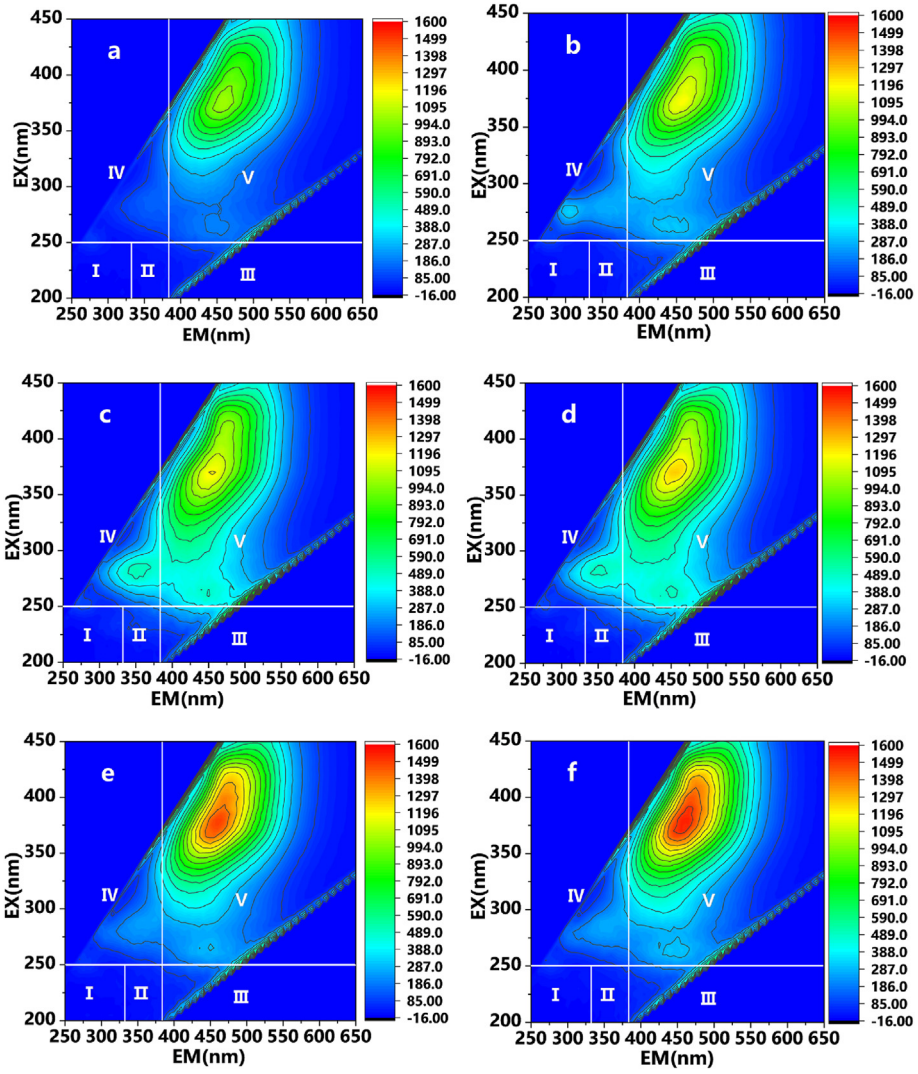


Fig. 3. EEM images of C1.1 (a), M1.1 (b), C1.5 (c), M1.5(d), C2.1 (e) and M2.1 (f). [tyrosine-like (I), Tryptophan-like (II), fulvic acid-like (III), soluble microbial byproduct-like (IV) and humic acid-like organics (V)].

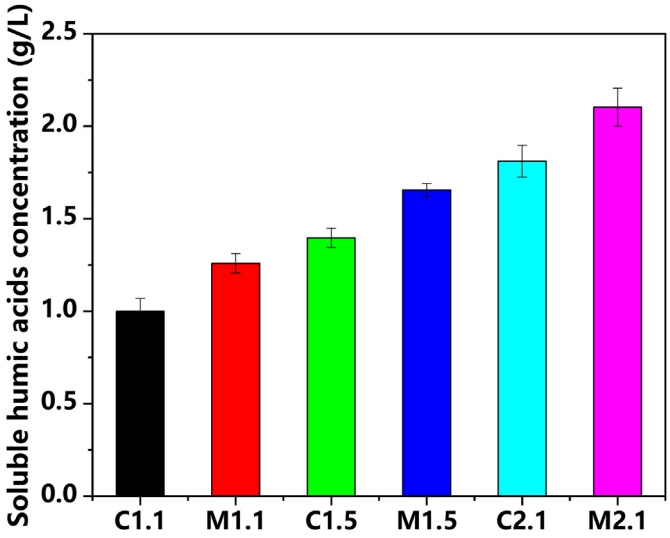


Fig. 4. Soluble humic acids concentrations in C1.1, M1.1, C1.5, M1.5, C2.1 and M2.1.

Table 1
Daily methane yield statistics from the final 30 days under different OLRs in control and micro-aeration group.

Groups	Average (NmL/d)	Max (NmL/d)	Min (NmL/d)
C1.1	441.5 ± 26.7	480.8 ± 10.0	391.9 ± 10.0
M1.1	468.0 ± 20.7	533.5 ± 23.6	394.1 ± 26.6
C1.5	470.9 ± 20.0	557.7 ± 4.8	381.0 ± 15.5
M1.5	517.8 ± 17.6	632.8 ± 21.7	432.5 ± 1.5.0
C2.1	629.6 ± 22.4	694.0 ± 7.5	549.8 ± 23.3
M2.1	674.5 ± 22.7	751.5 ± 5.8	612.8 ± 19.0

the breakdown of polysaccharides and ferment mono sugars into organic acids [12,30]. The relative abundances of *Cloacimonadia* (phylum *Cloacimonadota*) was low in the samples of C1.1, M1.1, C1.5 and M1.5, while it became the third dominant bacteria in C2.1 and M2.1 at the class level, which were 13.0% and 24.0% after running time of 238 days. *Cloacimonadia* are considered as syntrophic microorganism that have ability to convert propionate and butyrate into acetate with some microbes basing on their metagenomics sequence data [31,32]. Nevertheless, the cultivation attempts of *Cloacimonadia* were failed because of their requirements for

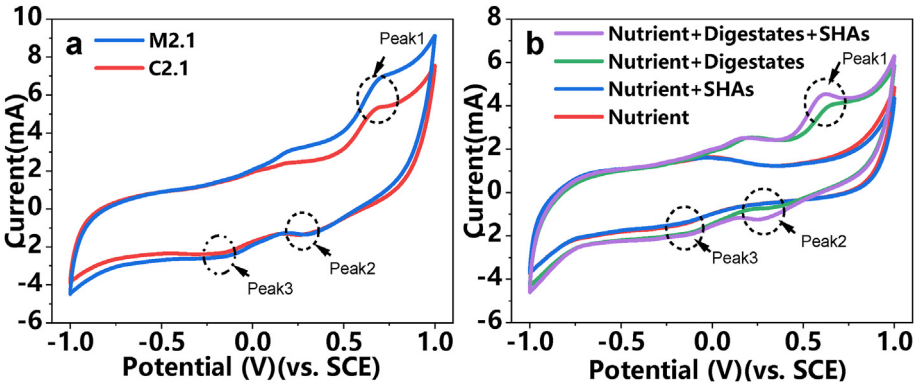


Fig. 5. Cyclic voltammetry of digestate (a) and digestate with SHAs (b).

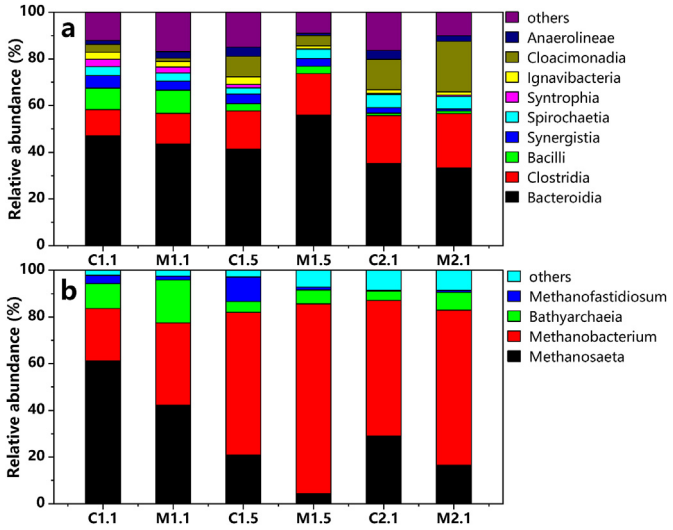


Fig. 6. Bacterial community structures at class level (a) and archaeal community structures at genus level (b) of C1.1, M1.1, C1.5, M1.5, C2.1 and M2.1. Class and genus level less than 1% of the total sequences and unclassified class were classified as others.

obligatory syntrophism with other microorganisms [31]. In this study, the relative abundance of *Cloacimonadia* increased obviously after a long-time cultivation, which could be further increased under micro-aeration condition. Similar results were proved by a recent study that the relative abundance of *Cloacimonadia* under sustainable supply of nano-oxygen bubbles increased from 1.8% to 23.7% compared with the control [33]. The bacterial communities were shifted by the OLRs and micro-aeration. In addition, the bacterial communities under micro-aeration conditions showed higher abilities of hydrolysis, homoacetogenic acetate production according to syntrophism.

The archaeal community structures of C1.1, M1.1, C1.5, M1.5, C2.1 and M2.1 at genus level are shown in Fig. 6b. *Methanosaeta* and *Methanobacterium* were the most two dominant genera in all collected samples. The relative abundances of *Methanosaeta*, a specialist of exclusively using acetate as substrate to producing methane [34], was at the dominant position in C1.1 and M1.1. However, the archaea at the dominant position of relative abundances in C1.5, M1.5, C2.1 and M2.1 were shifted to be *Methanobacterium*, which is reported to be the hydrogenotrophic archaea [35]. The relative abundances of *Methanosaeta* in M1.1, M1.5 and M2.1 were 42.2%, 4.2% and 16.4%, which were 68.9%, 20.3% and 57.0% of that in C1.1, C1.5, and C2.1, respectively. The results indicated that micro-aeration would inhibit the growth of *Methanosaeta*. Similar results were also reported by a previous study [14], who found the growth of *Methanosaeta* negatively affected by the oxygen dosages, and its abundance was remarkable lower at higher micro-aeration condition. The relative abundances of *Methanobacterium* in M1.1, M1.5 and M2.1 were 35.2%, 81.4% and 66.4%, which were 56.9%, 33.3% and 14.3% higher than those of C1.1, C1.5, and C2.1, respectively. *Methanobacterium* had higher abundances in all micro-aerobic samples (M1.1, M1.5 and M2.1), suggesting that hydrogenotrophic methanogens might possess a better oxygen tolerance and finally become the dominant pathway to produce methane under micro-aeration condition [14]. The analysis results mentioned above indicated that micro-aeration stimulated the growth of *Clostridia* and *Cloacimonadia* to promote the degradation of lignocellulosic biomass and VFAs syntrophic-oxidization, providing more substrates for methanogens. At the same time, oxygen-tolerant methanogens (e.g., *Methanobacterium*) became the domain genera under micro-aerobic conditions, making the methanogenesis on the rail.

3.6. Enzyme represented metabolic functions by PICRUSt

Hydrolysis is regarded as the rate-limiting step during AD of cellulosic waste e.g., corn straw, and micro-aeration treatment has been reported to be efficient in accelerating the hydrolysis process by stimulating the growth of facultative bacteria [5,14]. Biodegradation of cellulose involves a series of extracellular cellulases e.g., endoglucanase (EC.3.2.1.4) and beta-glucosidase (EC.3.2.1.21) [36]. The relative abundances of critical enzymes related to the lignocellulose degradation in C1.1, M1.1, C1.5, M1.5, C2.1 and M2.1 are shown in Fig. 7. Endoglucanase and beta-glucosidase in M1.1, M1.5 and M2.1 possessed higher relative abundance than those in C1.1, C1.5 and C2.1. Xylanase (EC 3.2.1.8) is an enzyme to degrade xylan, which is the main content of hemicellulose in corn straw. The relative abundance of xylanase in M1.1, M1.5 and M2.1 were all higher compared with the control, especially in M1.5, which was 163% of C1.5. The higher cellulases and xylanase activity predicted by PICRUSt in micro-aeration group indicated that micro-aeration could stimulate the excretion of extracellular enzymes, thus accelerate the degradation of lignocellulosic substrate, and benefit the methane production.

3.7. Effects of different concentration of SHAs on methanogenesis

Under micro-aeration conditions, the growth of facultative bacteria was stimulated (Fig. 6a), more cellulase and xylanase were excreted (Fig. 7), and SHAs concentrations were therefore higher (Figs. 3 and 4). In order to investigate the effect of extracted SHAs

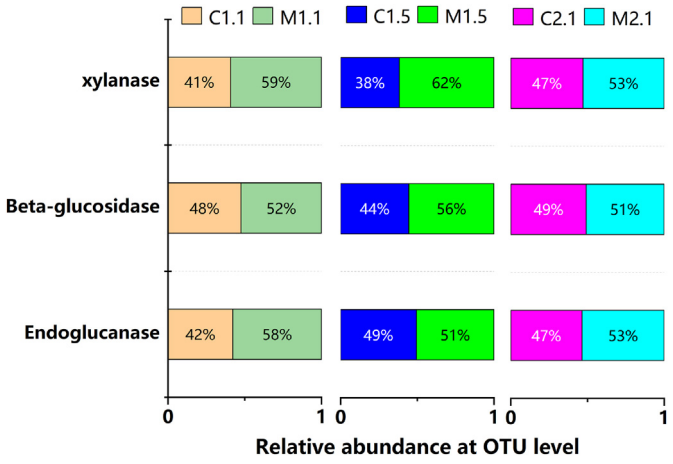


Fig. 7. Relative abundances of critical enzymes related to the degradation of lignocellulose in C1.1, M1.1, C1.5, M1.5, C2.1 and M2.1.

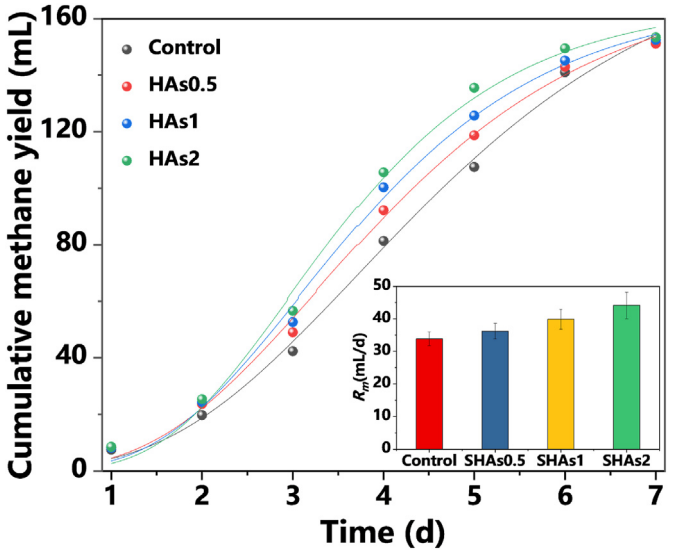


Fig. 8. Effects of extracted SHAs with different concentrations on methanogenesis.

on methanogenesis, SHAs were added into AD of sodium acetate at 4 g/L, and the results are shown in Fig. 8. SHAs concentrations had no effect on the cumulative methane yield but obviously promoted the methane production rate. For instances, the maximum methane production rates (the modified *Gompertz* model fitting results) of SHAs0.5, SHAs1 and SHAs2 were 36.2 ± 2.4, 39.9 ± 3.0 and 44.1 ± 4.1 mL/(g VS d), which were 6.9%,17.7% and 30.3% higher than those in control, respectively. Several studies also found the similar results that the methane production rate from AD of acetate could be promoted by the enhancement of DIET with the addition of HAs, magnetite, nanoFe_3O_4 or sludge-derived hydrochar [21,37–39]. Under the help of electron shuttles, the acetate was not directly converted into CH_4, but firstly syntrophic oxidized into CO_2, H^+ and e^- and then methanogens accepted e^- and H^+ to reduce CO_2 for methane production [21]. The results indicated that SHAs might also be used as a mediated material to facilitate the DIET between syntrophic bacteria e.g., *Synergistia* [40], *Spirochaetia* [41], *Syntrophia* [42], and *Cloacimonadia* [31,32] (Fig. 6a) and methanogens e.g. *Methanobacterium* [43] and *Methanosaeta* [44] (Fig. 6b), leading to higher methane production rate under micro-aerobic conditions.

4. Conclusion

Micro-aeration led enhanced methane yield during semi-continued AD of corn straw by regulation the microbial community structures. Facultative bacteria that are capable of hydrolytic enzyme secretion were stimulated by micro-aeration, leading to faster hydrolysis and higher SHAs production. Meanwhile, methanogens with a certain oxygen tolerance were enriched, making the methanogenesis on the rail. In addition, the produced SHAs could benefit the syntrophic methanation by possibility acting as mediator for DIET. The implementation of this study proved the beneficial effects of micro-aeration on the semi-continued AD of cellulosic substrate, which could also provide references for the industrial application of micro-aeration technology.

CRediT authorship contribution statement

Rong Zhu: Investigation, Methodology, Data curation, Writing – original draft. **Ya-feng Zhang:** Validation, Data curation. **Hua Zou:** Writing – review & editing, Resources, Supervision, Visualization. **Rong-Bo Guo:** Writing – review & editing. **Shan-Fei Fu:** Conceptualization, Methodology, Writing – review & editing, All authors read and approved the final manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.renene.2022.06.106.

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